

Yogesh Muppiri

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Professional Summary: Data and Machine Learning Engineer with 3+ years of experience designing scalable data pipelines, developing predictive analytics, and building business intelligence solutions on cloud-native architectures. Proven expertise in Python, SQL, AWS, Tableau, and ML frameworks to transform large-scale data into actionable strategies that drive product innovation and operational excellence.

EDUCATION

The University of Texas at Dallas

Master of Science, Business analytics and artificial intelligence

May 2024

GPA 3.65

GITAM University

Bachelor of Technology, Computer Science Engineering

May 2022

GPA 3.4

Relevant Coursework: Machine Learning, Big Data, Database Foundations, Natural Language Processing, Business Enterprise and Advanced Statistics, Predictive Analytics, Business Data Warehousing, Data Visualization and artificial intelligence.

TECHNICAL SKILLS

Programming & Cloud: Python, SQL, R, AWS, Azure

Machine learning and NLP: Predictive Modelling, Linear Regression, Logistic Regression, Decision Trees, K-Nearest Neighbors (KNN), Support Vector Machines (SVM), Random Forest, XGBoost, PyTorch, TensorFlow, Scikit-learn Matplotlib, NLTK, spaCy.

Data Analytics and Management: PostgreSQL, MySQL, Apache Airflow, PySpark, Hadoop, Dask, Tableau, Grafana, NumPy, Pandas, ETL (Extract, Transform, Load), Advance Statistics.

Other skills: Hadoop, hive, AWS EC2, S3, Amazon RDS, CI/CD, AWS Lambda, AWS SageMaker, Django, Hugging Face Transformers, Splunk, Git, GitHub, Kubernetes, BART, Stable Diffusion, Keras, Spark MLlib, API Integration & Development, React.js.

PROFESSIONAL EXPERIENCE

Community Dreams Foundation, Florida, USA

July 2024 – Present

Data Engineer

- Built and optimized scalable Spark, Kafka, and Airflow pipelines, enabling efficient processing of 1TB+ structured and unstructured data daily, reducing ETL latency by 20%.
- Engineered low-latency LLaMA-based inference APIs with FastAPI and TensorRT acceleration, achieving a 4x throughput increase and reducing latency to under 15ms, enabling real-time AI application.
- Maintained a scalable data lake using AWS Lake Formation, Glue, and S3 to manage 5TB+ of multi-format datasets, ensuring encryption, role-based access control, and optimized Redshift Spectrum integration.
- Collaborated on automating AWS infrastructure provisioning via Terraform, contributing to setup of key resources like S3 buckets, IAM roles, and VPCs.

Tekflaire Inc, Texas, USA

May 2023 – June 2024

Machine Learning Engineer

- Developed real-time personalization engines using Neural Collaborative Filtering and BERT4Rec for E-commerce platform, enhancing recommendation precision for over 500K active users.
- Spearheaded a multilingual NLP pipeline using Hugging Face Transformers, processing 800k+ customer records across 3 languages: improved entity extraction and sentiment classification accuracy by 18%.
- Architected scalable ML pipelines using AWS Lambda and Step Functions for SageMaker batch transforms, reducing job execution time by 35% and lowering infrastructure costs by 20%.
- Formulated advanced churn prediction models leveraging XGBoost, Random Forest, and Deep Learning; achieved a 15% decrease in churn rate and enabled sub-500ms fraud detection via Kafka-based inference.

HDLC Technologies, Chennai, India

December 2020 – April 2022

Data analyst

- Initiated CI/CD pipelines using Jenkins and GitHub Actions for automated data deployments, reducing manual errors by 40% and improving pipeline reliability.
- Created real-time dashboards in Power BI and Tableau, sourcing data from Apache Kafka streams to enable up-to-date decision-making across business units. Automated A/B testing workflows using Apache Airflow, Spark, and SQL, reducing analysis turnaround time by 40% and enabling real-time campaign optimization.
- Established serverless infrastructure by configuring AWS Lambda, API Gateway, and Auto Scaling Groups, improving system latency by 20% and enhancing scalability and cost-efficiency.
- Conducted advanced cohort and funnel analyses using SQL and Python, identifying user behaviour trends that led to a 12% increase in customer retention and informed data-backed product strategy improvements.

ACADEMIC PROJECT EXPERIENCE

AI Text to Video generator using stable diffusion models

November 2023 – December 2023

- Devised an AI video generation system using Stable Diffusion for visuals and gTTS for narration, reducing content creation time by 25%.
- Implemented MoviePy to synchronize visuals, text overlays, and audio, boosting multimedia output efficiency by 30%.
- Architected a scalable video pipeline on AWS, leveraging S3 for storage and refining processing for concise, high-quality 1-minute videos.

Walmart Sales Forecasting & Analysis

January 2024 – March 2024

- Prototyped a Streamlit-based application to forecast weekly Walmart store sales using real-time inputs; enabled input of Store ID, Department, and Holiday, and visualized 3 years of sales trends across 45 stores, highlighting holiday impact and temperature effects through interactive heatmaps and time-series charts—boosting forecast interpretability by 30%.
- Designed and initiated a predictive modelling pipeline using ML models like Random Forest (92% accuracy) to forecast weekly sales across 45 Walmart stores based on temperature, fuel price, unemployment, holidays, and store attributes.

Publications: *Detection of Cyber-attacks using Machine Learning techniques.* ([view publication](#))